

AI Healthcare Information Assistant

Project Overview

The AI Healthcare Information Assistant is a Streamlit-based RAG (Retrieval-Augmented Generation) application that allows users to upload healthcare PDF documents and ask questions about their contents.

Key Features

- PDF Upload and Processing
- Text Extraction and Chunking
- Embeddings using Sentence Transformers
- FAISS Vector Database
- Question Answering over Documents
- Local AI Inference with Ollama
- BYOK Support (OpenAI/Gemini API Keys)

Workflow

PDF Upload → Text Extraction → Text Chunking → Embeddings → FAISS Vector Store → User Question → Retriever → Ollama (Local AI) or OpenAI/Gemini (BYOK) → Answer

Technology Stack

Streamlit, LangChain, FAISS, Sentence Transformers, Ollama, OpenAI API, Gemini API, PyPDF.

Step-by-Step Procedure

1. Create project folder and virtual environment.
2. Install required packages.
3. Create app.py and data folder.
4. Upload healthcare PDFs.
5. Extract text and split into chunks.
6. Generate embeddings and store in FAISS.
7. Build retriever and question-answering pipeline.
8. Integrate Ollama and BYOK options.
9. Run the Streamlit application.
10. Test and push the project to GitHub.